Lab Assignment 3: GridWorld – Policy Evaluation and Value Iteration

Task 1: Formulating the GridWorld Environment

For the **GridWorld environment**, the problem can be represented using the following Reinforcement Learning (RL) components:

RL Component	Description
States	Each cell in the GridWorld represents a state. The agent observes its current position in the grid and uses this information to decide its next action.
Actions	The agent can move **Up, Down, Left, or Right** from its current position. Actions that move the agent outside the grid or into restricted cells are not allowed.
Reward Structure	The agent receives a **positive reward** for reaching the goal and a **negative reward/penalty** for undesirable actions such as hitting obstacles or taking unnecessary steps. A small step penalty can also encourage the agent to find a shorter path.
Terminal State(s)	The **goal cell** is the main terminal state. When the agent reaches this state, the episode ends. If the environment contains a failure/trap state, reaching it can also terminate the episode.

Goal of the Agent	The agent's objective is to learn a policy that takes it from the starting position to the goal while maximizing the total cumulative reward and avoiding obstacles or unnecessary movements.

Task 2: Policy Evaluation

Policy evaluation repeatedly updates the value of each state according to the current policy until the values become stable.

Iteration	Maximum Change (Δ)	Convergence Status
1	**1.0000**	Not Converged
2	**0.9000**	Not Converged
3	**0.8100**	Not Converged
4	**0.7290**	Not Converged
5	**0.6561**	Not Converged

Task 3: Value Iteration and Optimal Policy Generation

Value iteration determines the maximum expected value for each state and selects the action that provides the best long-term return.

State	Optimal Action	State Value
S1	DOWN	-4.0951
S2	DOWN	-3.4390
S3	DOWN	-2.7100
S4	DOWN	-1.9000
S5	DOWN	-3.4390

Task 4: Optimal Path Analysis

Policy Type	Path Length	Total Reward	Goal Reached
Random Policy	20	-20.0	**No**
Optimal Policy	6	-5.0	**Yes**
Policy Type	**Path Length**	**Total Reward**	**Goal Reached**

Analysis

The **Random Policy** takes a longer and less efficient route because actions are selected without considering their expected rewards. The **Evaluated Policy** performs better because it uses the state values learned during policy evaluation. The **Optimal Policy** produces the shortest path and highest total reward by selecting the best action at every state.

Overall Conclusion

The GridWorld experiment demonstrates how reinforcement learning improves an agent's decision-making through policy evaluation and value iteration. Policy evaluation estimates the usefulness of states under a given policy, while value iteration identifies the best actions for maximizing future rewards. The optimal policy results in a shorter path and higher cumulative reward compared with random and evaluated policies.

Analysis Questions

Q1. What is a state in GridWorld?

Answer:
A state represents the **current position of the agent in the GridWorld**. In a 4×4 GridWorld, there are 16 possible states, represented as **S1 to S16**, with **S16 as the goal state**.

Q2. What actions can the agent perform?

Answer:
The agent can perform four actions: **UP, DOWN, LEFT, and RIGHT**. These actions move the agent from its current cell to an adjacent cell.

Q3. What is the purpose of Policy Evaluation?

Answer:
The purpose of Policy Evaluation is to **estimate the value of each state under a given policy**. It repeatedly updates the state values until they become stable or reach the specified number of iterations.

Q4. How does Value Iteration differ from Policy Evaluation?

Answer:
Policy Evaluation calculates the value of states for a **fixed policy**, whereas **Value Iteration** directly searches for the **best action** at each state by selecting the action with the maximum expected value. Therefore, Value Iteration can generate the optimal policy.

Q5. What is the Bellman Optimality Equation?

Answer:
The Bellman Optimality Equation gives the maximum expected value of a state by choosing the best available action:

$V*(s)=maxa[R(s,a)+\gamma V*(s')]$V^*(s)=\max_a \left[R(s,a)+\gamma V^*(s')\right]

It is used in Value Iteration to determine the **optimal value and optimal action** for each state.

Q6. Which policy reached the goal using the fewest steps?

Answer:
The **Optimal Policy** reached the goal using the fewest steps. In the experiment, it reached the goal in **6 steps**, following the path:

S1 → S5 → S9 → S13 → S14 → S15 → S16

Q7. Why is the optimal policy superior to a random policy?

Answer:
The optimal policy selects actions based on the **learned state values**, allowing the agent to move toward the goal efficiently. A random policy selects actions without considering future rewards, which can result in unnecessary movements, longer paths, and failure to reach the goal.

Q8. How would obstacles affect the optimal path?

Answer:
Obstacles would prevent the agent from moving through certain cells. Therefore, the optimal policy would need to find an **alternative path around the obstacles**. Depending on their location, obstacles could increase the path length and reduce the total reward received by the agent.